Aalto University
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Brief summary on excited states

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Excited states

- Photochemistry and photophysics of molecules and materials are highly active fields of research.
- Many important technologies such as light-emitting diodes and solar cells depend on the behavior of excited states.
- Excited states can be studied for example with Time-Dependent Density Functional Theory (TD-DFT). This is also the method primarily used in the present tutorial.
  - A review "Density functional methods for excited states: equilibrium structure and electronic spectra" from Furche and Rappoport is openly available at https://escholarship.org/uc/item/7z63q82s

Figure: Wikimedia Commons / PiccoloNamek (CC BY-SA)

Electromagnetic spectrum visible to the human eye

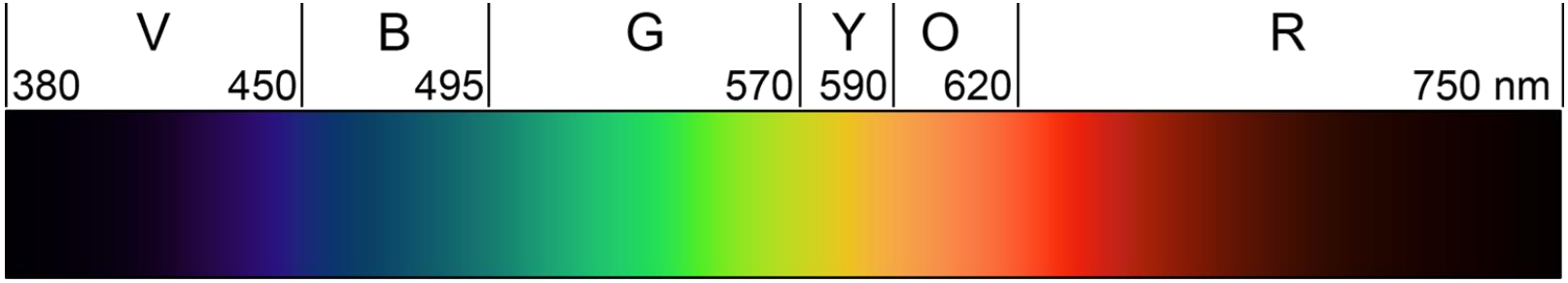


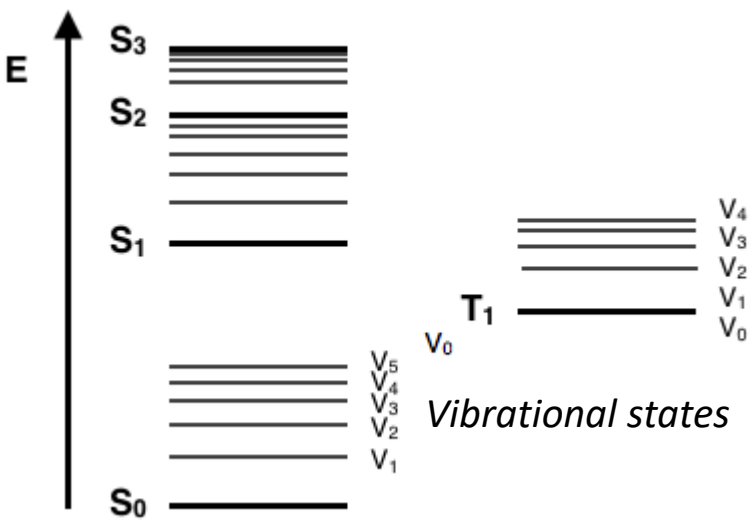
Figure: Wikimedia Commons (Public Domain)

Color	Wavelength (nm)	Frequency (THz)	Photon energy (eV)
violet	380–450	670–790	2.75–3.26
blue	450–485	620–670	2.56–2.75
cyan	485–500	600–620	2.48–2.56
green	500–565	530–600	2.19–2.48
yellow	565–590	510–530	2.10–2.19
orange	590–625	480–510	1.98–2.10
red	625–750	400–480	1.65–1.98

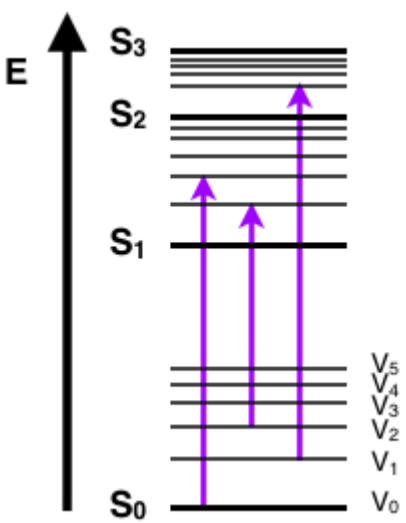
https://en.wikipedia.org/wiki/Visible_spectrum

Jablonski diagrams (1/3)

Excited states of molecules are typically schematically illustrated with Jablonski diagrams that show the electronic and vibrational excited states.



Foundation of a typical Jablonski Diagram



Three possible absorption transitions represented.
These can be measured by **UV/Vis** spectroscopy. **The tutorial focuses on this kind of transitions**.

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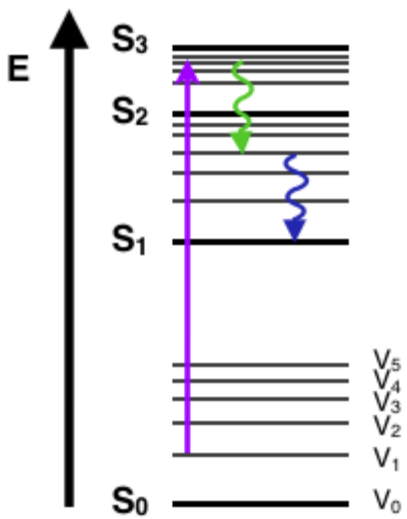
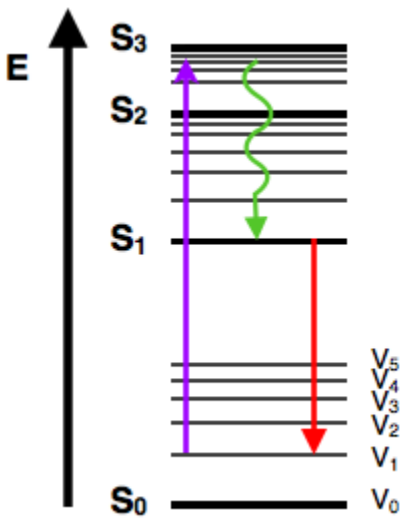
Jablonski diagrams (2/3)



Possible scenario with absorption, internal conversion, and vibrational relaxation *processes shown.*



Possible scenario with absorption, internal conversion and vibrational relaxation, and fluorescence processes shown.

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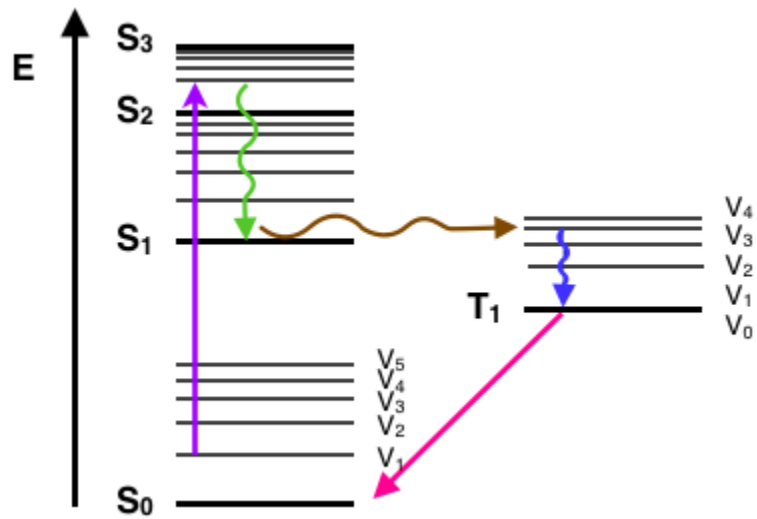
Jablonski diagrams (3/3)



Possible scenario with absorption, internal conversion, vibrational relaxation, intersystem crossing, and phosphorescence processes shown.

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Timescales

These are only for information. Our TD-DFT calculations are in reality time-independent.

Transition	Timescale	Radiative Process?
Internal Conversion	10^{-14} - 10^{-11} s	no
Vibrational Relaxation	10^{-14} - 10^{-11} s	no
Absorption	10^{-15} s	yes
Phosphorescence	10^{-4} - 10^{-1} s	yes
Intersystem Crossing	10^{-8} - 10^{-3} s	no
Fluorescence	10^{-9} - 10^{-7} s	yes

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